A CLASSIFICATION OF QUBIT ISOMETRIES FOR THE ONE-OBSERVABLE WASSERSTEIN DIVERGENCE

AGENT LANE 15

Abstract. Simon and Virosztek asked for the structure of qubit state-space isometries with respect to the quantum Wasserstein divergences d_{xz} and d_z. This note settles the d_z part. A self-map of the qubit state space is a d_z-isometry if and only if it preserves the Euclidean length of every Bloch vector and globally either fixes the z-coordinate or changes its sign. Thus the d_z-isometries have the same Bloch-coordinate classification as the D_z-isometries in the source paper.

1. Source question

The source paper is Richard Simon and Daniel Virosztek, *Isometries of the qubit state space with respect to quantum Wasserstein distances*, arXiv:2408.09879. In its final remarks the authors ask for the structure of qubit state-space isometries for the quantum Wasserstein divergences d_{xz} and d_z. The present note answers the d_z question only.

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> **Final remarks.** Given the existing results on the isometries of the quantum bit state space with respect to the quantum Wasserstein distances D_{sym} and D_{xz} (see [23]), and also our present results on the preservers of the divergence d_{sym} and the distance D_z, it is highly natural to ask what can be said about the structure of the qubit state space isometries with respect to the quantum Wasserstein divergences d_{xz} and d_z. We believe that these questions are highly nontrivial — especially as quantum Wasserstein *divergences* are typically non-convex (in contract to quantum Wasserstein *distances*), and hence it is often the case that their extremal values (maxima) are not realaized by extremal (that is: pure) states —, and we consider them as attractive open problems.
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2. Notation

Write each qubit state as

$$\rho_{\mathbf{b}} = \frac{1}{2}(I + x\sigma_x + y\sigma_y + z\sigma_z), \qquad \mathbf{b} = (x, y, z) \in \mathbb{R}^3, \qquad |\mathbf{b}| \le 1.$$

Let $p_t = (I + t\sigma_z)/2$ for $t = \pm 1$. The one-observable cost is

$$C_z = (\sigma_z \otimes I - I \otimes \sigma_z^T)^2 = \mathrm{diag}(0, 4, 4, 0),$$

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in the standard ordered basis. Let D_z be the corresponding quantum Wasserstein distance and

$$d_z^2(\rho,\omega) = D_z^2(\rho,\omega) - \frac{1}{2}D_z^2(\rho,\rho) - \frac{1}{2}D_z^2(\omega,\omega)$$

be the associated divergence.

3. Elementary formulas

Lemma 1 (Self-distance and pole distances)**.** *Let* $\rho = \rho_{\mathbf{b}}$ *with* $\mathbf{b} = (x,y,z)$ *and* $r = |\mathbf{b}|$. *Set*

$$F_z(\rho) := D_z^2(\rho,\rho).$$

Then

$$F_z(\rho) = \frac{2(x^2+y^2)}{1+\sqrt{1-r^2}},$$

with the value 0 *at* $r=0$. *Moreover, for* $t = \pm 1$,

$$D_z^2(\rho,p_t) = 2 - 2tz, \qquad d_z^2(\rho,p_t) = 2 - 2tz - \frac{1}{2}F_z(\rho).$$

Proof. If the second marginal is the pure state p_t, then every coupling is the product coupling $p_t \otimes \rho^T$. Evaluating $C_z = 2I\otimes I - 2\sigma_z \otimes \sigma_z^T$ on this coupling gives

$$D_z^2(\rho,p_t) = 2 - 2tz.$$

Since $D_z^2(p_t,p_t) = 0$, the displayed formula for $d_z^2(\rho,p_t)$ follows once the self-distance formula is known.

For the self-distance, use the canonical-purification formula recalled in the source paper:

$$D_z^2(\rho,\rho) = \langle\!\langle \sqrt{\rho} \| \, C_z \, \| \sqrt{\rho} \rangle\!\rangle.$$

The spectral form of the cost is

$$C_z = 2\,\|\sigma_x\rangle\!\rangle\langle\!\langle\sigma_x\| + 2\,\|\sigma_y\rangle\!\rangle\langle\!\langle\sigma_y\|.$$

For $r > 0$,

$$\sqrt{\rho} = aI + b\,\frac{x\sigma_x + y\sigma_y + z\sigma_z}{r}, \qquad b^2 = \frac{1-\sqrt{1-r^2}}{4}.$$

Using $\operatorname{tr}(\sigma_i\sigma_j) = 2\delta_{ij}$, we obtain

$$D_z^2(\rho,\rho) = 2\left(\operatorname{tr}(\sigma_x\sqrt{\rho})^2 + \operatorname{tr}(\sigma_y\sqrt{\rho})^2\right) = 8b^2\,\frac{x^2+y^2}{r^2}.$$

This equals

$$2\left(1-\sqrt{1-r^2}\right)\frac{x^2+y^2}{r^2} = \frac{2(x^2+y^2)}{1+\sqrt{1-r^2}}.$$

The final expression extends continuously to $r=0$, giving $F_z(I/2) = 0$. $\square$

Lemma 2 (Strict recovery of the Bloch radius)**.** *Fix $z \in [-1,1]$. The function*

$$r \longmapsto \frac{2(r^2-z^2)}{1+\sqrt{1-r^2}}, \qquad |z| \le r \le 1,$$

is strictly increasing on its nontrivial domain.

Proof. If $|z|=1$ the domain is a singleton. Otherwise put $s=\sqrt{1-r^2}$. For $|z|<r<1$,

$$\frac{d}{dr}\left(\frac{r^2-z^2}{1+s}\right)=\frac{r}{s}\left(1-\frac{z^2}{(1+s)^2}\right)>0.$$

Indeed $z^2 \le r^2 = 1-s^2 < (1+s)^2$. Continuity gives strict increase on the whole interval. □

Lemma 3 (The d_z-diameter pair)**.** *The d_z-diameter of $\mathcal{S}(\mathbb{C}^2)$ is 2, and the only unordered pair realizing it is $\{p_+, p_-\}$.*

Proof. The trivial coupling bound used in the source proof of Theorem 2 gives

$$D_z^2(\rho_{\mathbf{b}}, \rho_{\mathbf{b}'}) \le 2 - 2zz' \le 4.$$

Because self-distances are nonnegative, $d_z^2(\rho,\omega) \le D_z^2(\rho,\omega)$, so $d_z \le 2$. The pole pair satisfies

$$d_z(p_+,p_-) = D_z(p_+,p_-) = 2.$$

If $d_z(\rho,\omega)=2$, then $D_z(\rho,\omega)=2$ as well. The equality case in the same trivial-coupling bound forces $z=1$ and $z'=-1$, or conversely; hence the states are exactly p_+ and p_-. □

4. Idea of the proof

The divergence removes self-distances, so a d_z-isometry does not visibly preserve $D_z(\rho,\rho)$. The pole states restore that missing information. For an arbitrary state ρ, the difference of the squared distances from p_+ and p_- recovers the z-coordinate, while the sum recovers the self-distance $F_z(\rho) = D_z^2(\rho,\rho)$. Since F_z is strictly monotone in the Bloch radius once z is fixed, the divergence itself forces the same invariants that Simon and Virosztek proved characterize the D_z-isometries.

5. Classification theorem

Theorem 1. *Let $\Phi : \mathcal{S}(\mathbb{C}^2) \to \mathcal{S}(\mathbb{C}^2)$ be any map. Then the following are equivalent.*

(1) *Φ is a d_z-isometry:*

$$d_z(\Phi(\rho),\Phi(\omega)) = d_z(\rho,\omega) \qquad (\rho,\omega \in \mathcal{S}(\mathbb{C}^2)).$$

(2) *Φ preserves Bloch radii and globally either fixes or flips the z-coordinate:*

$$|\mathbf{b}_{\Phi(\rho)}| = |\mathbf{b}_\rho| \quad (\rho \in \mathcal{S}(\mathbb{C}^2)),$$

and either

$$(\mathbf{b}_{\Phi(\rho)})_3 = (\mathbf{b}_\rho)_3 \quad (\rho \in \mathcal{S}(\mathbb{C}^2)),$$

or

$$(\mathbf{b}_{\Phi(\rho)})_3 = -(\mathbf{b}_\rho)_3 \quad (\rho \in \mathcal{S}(\mathbb{C}^2)).$$

Proof. Assume first that Φ preserves d_z. By Lemma 3, Φ maps the pole pair onto itself. Therefore there is a sign $\varepsilon \in \{1, -1\}$ such that

$$\Phi(p_t) = p_{\varepsilon t} \qquad (t = \pm 1).$$

Let $\rho = \rho_{\mathbf{b}}$, let $\Phi(\rho) = \rho_{\mathbf{b}'}$, and write

$$z = (\mathbf{b})_3, \qquad z' = (\mathbf{b}')_3, \qquad F = F_z(\rho), \qquad F' = F_z(\Phi(\rho)).$$

For $t = \pm 1$, distance preservation and Lemma 1 give

$$2 - 2\varepsilon t z' - \frac{1}{2}F' = d_z^2(\Phi(\rho), p_{\varepsilon t}) = d_z^2(\rho, p_t) = 2 - 2tz - \frac{1}{2}F.$$

Subtracting the equations for $t = 1$ and $t = -1$ yields $z' = \varepsilon z$. Adding them yields $F' = F$.

Now $|z'| = |z|$, and

$$F_z(\rho_{\mathbf{b}}) = \frac{2(|\mathbf{b}|^2 - z^2)}{1 + \sqrt{1 - |\mathbf{b}|^2}}.$$

Lemma 2 therefore implies $|\mathbf{b}'| = |\mathbf{b}|$. Hence the Bloch radius is preserved and the z-coordinate is globally fixed if $\varepsilon = 1$, globally flipped if $\varepsilon = -1$.

Conversely, assume the Bloch-coordinate condition. Theorem 2 of Simon and Virosztek states precisely that this condition is equivalent to preservation of the one-observable Wasserstein distance D_z. Thus

$$D_z(\Phi(\rho), \Phi(\omega)) = D_z(\rho, \omega) \qquad (\rho, \omega \in \mathcal{S}(\mathbb{C}^2)).$$

By Lemma 1, $F_z(\rho)$ depends only on $|\mathbf{b}_\rho|$ and $(\mathbf{b}_\rho)_3^2$. The assumed Bloch-coordinate condition therefore also preserves $F_z(\rho) = D_z^2(\rho, \rho)$. Substituting these two invariances into the definition of d_z gives

$$d_z(\Phi(\rho), \Phi(\omega)) = d_z(\rho, \omega)$$

for all states ρ, ω. This proves the converse. $\square$

6. Verification and novelty notes

The numerical script `tmp/verify_dz_formulas.py` checks the self-distance identity in Lemma 1 by explicit vectorization against $C_z = \mathrm{diag}(0, 4, 4, 0)$, including random Bloch vectors, the pole states, and pure equatorial states. It also checks the strict monotonicity used in Lemma 2 on representative grids.

Local run indexes were searched for arXiv:2408.09879 and core phrases around d_z, d_{xz}, Wasserstein divergences, and qubit isometries. No prior packet for this result was found. A bounded external search on 2026-07-19 found the source paper and a close June 2026 paper of Szabo and Virosztek on single-observable quantum Wasserstein isometries; the latter concerns distances D_A, not divergence isometries d_A, and so does not appear to answer the d_z open question settled here.

References

[1] R. Simon and D. Virosztek, *Isometries of the qubit state space with respect to quantum Wasserstein distances*, arXiv:2408.09879.

[2] G. De Palma and D. Trevisan, *Quantum optimal transport with quantum channels*, Ann. Henri Poincare 22 (2021), 3199–3234.

[3] E. Szabo and D. Virosztek, *Quantum Wasserstein isometries with a single observable generating the transport cost*, Acta Scientiarum Mathematicarum, published 27 June 2026. doi:10.1007/s44146-026-00254-5.